

Jialiang Gu

Ph.D. Student in Computer Science

✉ jgu7@gmu.edu

☎ (571) 234-2933

🌐 [gujialiang123.github.io](https://github.com/gujialiang123)

📍 Fairfax, VA, USA

Summary

- First-year Ph.D. student in Computer Science at George Mason University, advised by Prof. Keren Zhou. I also collaborate closely with Prof. Jialu Zhang at the University of Waterloo. My research interests focus on machine learning systems, LLM applications, and teaching and learning in the LLM era.

Education

Ph.D. Student	George Mason University , Computer Science • Advisor: Prof. Keren Zhou	Fairfax, VA, USA Jan 2025 – present
Undergraduate	Wuhan University , Computer Science, Hongyi Honor College • Overall GPA: 3.72/4.0	Wuhan, China Jan 2021 – Jan 2025

Published / Accepted Papers

NKI Teaching Lab: An Accelerator-Free Toolchain for Learning Tile-Level AI Accelerator Programming Jialiang Gu, Junjie Ma, Thai-Hoa Nguyen, Hao Wu, Zhicong Lu, Shuyin Jiao, Binqian Yin, and Keren Zhou. Accepted to SIGCSE TS 2027. Conference	2027
Exploring Cross-Scenario Generality of Agentic Memory Systems: Diagnostics and a Strong Baseline Zhikai Chen*, Jialiang Gu*, Junyu Yin*, Xianxuan Long, Shenglai Zeng, Xiaoze Liu, Kai Guo, Keren Zhou, and Jiliang Tang. Accepted to AACL-IJCNLP 2026. * Equal contribution (co-first authors). arXiv	2026
TileBench: A Controlled Benchmark for Performance Evaluation and Bottleneck Diagnosis of Tile-Based Programming Models Bowen Cui, Zhongchun Zhou, Hao Wu, Tejas Ramesh, Junyu Yin, Jialiang Gu, and Keren Zhou. Accepted to AACL-IJCNLP 2026. arXiv	2026
FrontierCS: Evolving Challenges for Evolving Intelligence Qiuyang Mang*, Wenhao Chai*, Zhifei Li*, Huanzhi Mao*, Shang Zhou*, Alexander Du*, Hanchen Li*, Shu Liu*, Edwin Chen, Yichuan Wang, Xieting Chu, Zerui Cheng, Yuan Xu, Tian Xia, Zirui Wang, Tianneng Shi, Jianzhu Yao, Yilong Zhao, Qizheng Zhang, Charlie F. Ruan, Zeyu Shen, Kaiyuan Liu, Zhaoyang Hong, Alex Gu, Ziyi Zhang, Runyuan He, Dong Xing, Zerui Li, Zirong Zeng, Yige Jiang, Lufeng Cheng, Ziyi Zhao, Youran Sun, Suyang Zhong, Junpeng Wang, Donglin Li, Wenyuan Huang, Jialiang Gu, Wesley Kai Zheng, Wangmeiyu Zhang, Ruyi Ji, Xuechang Tu, Zihan Zheng, Zhaozi Wang, Zexing Chen, Jingbang Chen, Jialu Zhang, Aleksandra Korolova, Peter Henderson, Pramod Viswanath, Vijay Ganesh, Saining Xie, Zhuang Liu, Dawn Song, Sewon Min, Ion Stoica, Joseph E. Gonzalez, Jingbo Shang, and Alvin Cheung. ICML, 2026. * Equal contribution (co-first authors). Paper · Code · Website	2026
Context-Aware Feedback Compression in Online Judge Programming with LLMs Jialiang Gu, Keren Zhou, Daming Li, Hanyuan Shi, and Jialu Zhang. FSE-IVR, 2026.	2026

arXiv / Manuscripts

A Systematic Study of Time Limit Exceeded Errors in Online Programming Assignments

2025

Jialiang Gu*, Jialu Zhang*, Wangmeiyu Zhang, José Pablo Cambronero, John Kolesar, Ruzica Piskac, Daming Li, and Hanyuan Shi. arXiv:2510.14339 [cs.SE], 2025. * Equal contribution (co-first authors).
[arXiv](#)

DiscoSum: Discourse-aware News Summarization

2025

Jia-Liang Gu*, Tenghao Huang*, Alexander Spangher*, Jiatong Shi, and Muhao Chen. arXiv:2506.06930 [cs.SE], 2025. * Equal contribution (co-first authors).
[arXiv](#)

Work Experience

Microsoft Research, Research Intern, AIFX Group

Redmond, WA, USA

Worked on regime-aware LLM inference optimization, improving LLM serving performance across diverse workload patterns on NVIDIA H200 GPUs.

May 2026 – Aug 2026

- Built an automated profiling and tuning pipeline for SGLang, exploring server-level configurations across long-context and concurrent workloads, including scheduling, chunked prefill, CUDA graphs, memory allocation, and attention/MoE backends.
- Achieved up to **2.2× end-to-end speedup** over the baseline in long-context regimes through workload-specific server optimization.
- Investigated kernel-level performance bottlenecks in prefill and decoding, using GPU profiling, correctness validation, and targeted kernel optimizations to reduce memory movement and improve execution efficiency.
- Designed extensive ablation and transferability studies to understand when optimizations generalize across workloads, including synthetic regimes and agentic workloads such as ToolAgent.
- Explored a workload-aware multi-server routing design that dispatches requests to specialized server configurations based on workload characteristics.
- Contributed performance findings, bug reports, and fixes upstream to the SGLang open-source project.

Research Experience

University of Waterloo, Research Assistant

Waterloo, Canada

Study on Time Limit Exceeded errors characteristics in competitive programming.

Jan 2023 – Jan 2025

- Advisor: Prof. Jialu Zhang

University of California, Davis, Research Intern

Davis, CA, USA

Structural summarization with large language models.

July 2024 – Sept 2024

- Advisor: Prof. Muhao Chen

Academic Competition Awards

Champion, 2024 ICPC China Hubei Provincial Contest (Wuhan)

May 2024

Gold Medal, International Collegiate Programming Contest (ACM-ICPC) Hangzhou Regional Contest

Dec 2023

Bronze Medal, International Collegiate Programming Contest (ACM-ICPC) EC-Final

Jan 2024

Silver Medal, International Collegiate Programming Contest (ACM-ICPC) Macau Regional Contest

Nov 2023

Silver Medal, International Collegiate Programming Contest (ACM-ICPC) Xi'an Regional Contest

Nov 2022

Silver Medal, International Collegiate Programming Contest (ACM-ICPC) Nanjing Regional Contest

Nov 2022

Gold Medal, 2023 ICPC China Shaanxi National Invitational Programming Contest (Xi'an)	May 2023
Gold Medal, 2023 CCPC China Hunan National Invitational Programming Contest (Xiangtan)	May 2023
Silver Medal, The 9th China Collegiate Programming Contest (Shenzhen)	Nov 2023
Silver Medal, The 8th China Collegiate Programming Contest (Weihai)	Nov 2022
First Prize, 2023 RAICOM Robotics Challenge national finals	Aug 2023
Third Prize, The 2nd CCF Siannan Cup Quantum Computing Challenge	Aug 2023

Honors and Experience at School

Excellent Teaching Assistant, Computer Composition and Design	Jan 2024
Excellent Student, Hongyi Honor College, Wuhan University (Top 15%)	Jan 2024
CCF Excellent Student, CCF National Outstanding College Student Award	Oct 2024
Assistant Head Teacher, Hongyi Honor College, Wuhan University	Jan 2022

Technical Skills

Programming Languages: C/C++, Python

Platforms: Windows, macOS, Linux

Other: Verilog, PyTorch

Certification: China Computer Federation Certified Software Professional (CCF-CSP): 470/500, Top 0.1%

Projects

Triton-Viz Core developer of an open-source project. Developed and maintained the visualizer module, focusing on extending Triton-Viz to support additional tile-level languages and improving its visualization infrastructure.	Jan 2025 – present
32-bit Pipelined CPU based on RISC-V Verilog personal project, Wuhan. Implemented a 32-bit pipelined RISC-V CPU in Verilog, supporting 32 basic instructions and successfully running C/C++-compiled programs and an FPGA-deployed aircraft shooting game.	July 2023 – present
Tripo APP Test engineer and project manager for a large-scale software application design course project. Built a travel diary app with post sharing, GPT-based summary generation, and AI-powered hotspot recommendation features.	Oct 2023 – present

Research Interests

- Machine learning systems (MLSys)
- LLM applications for code understanding, analysis, and explanation
- Ecosystem tools for next-generation GPU programming languages such as Triton
- LLM reasoning and reliability with static analysis and agent-based methods

Others

- Codeforces homepage: <https://codeforces.com/profile/gujialiang123>
- Language: English (CET-4 562)